

$$(i) \int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$$

$$\int e^x dx = e^x$$

$$(ii) \int \sin x dx = -\cos x$$

$$\int \tan x dx = -\log \cos x$$

$$\int \sec x dx = \log (\sec x + \tan x)$$

$$\int \sec^2 x dx = \tan x$$

$$(iii) \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \frac{a+x}{a-x}$$

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \frac{x-a}{x+a}$$

$$(iv) \int \sqrt{a^2 - x^2} dx = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$$

$$\int \sqrt{a^2 + x^2} dx = \frac{x\sqrt{a^2 + x^2}}{2} + \frac{a^2}{2} \sinh^{-1} \frac{x}{a}$$

$$\int \sqrt{x^2 - a^2} dx = \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \cosh^{-1} \frac{x}{a}$$

$$(v) \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$$

$$(vi) \int_0^\infty e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}$$

$$\int_0^\infty e^{-ax} \cos bx dx = \frac{a}{a^2 + b^2}$$

$$(vii) \int \sinh x dx = \cosh x$$

$$\int \cosh x dx = \sinh x$$

$$\int \tanh x dx = \log \cosh x$$

$$\int \coth x dx = \log \sinh x$$

$$\int \operatorname{sech}^2 x dx = \tanh x$$

$$\int \operatorname{cosech}^2 x dx = -\coth x$$

$$(viii) \int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \cos^n x dx = \frac{(n-1)(n-3)(n-5)\dots}{n(n-2)(n-4)\dots} \times \left(\frac{\pi}{2}, \text{ only if } n \text{ is even}\right)$$

$$\int_0^{\pi/2} \sin^m x \cos^n x dx = \frac{(m-1)(m-3)\dots \times (n-1)(n-3)\dots}{(m+n)(m+n-2)(m+n-4)\dots} \times \left(\frac{\pi}{2}, \text{ only if both } m \text{ and } n \text{ are even}\right)$$

$$(ix) \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx,$$

if $f(x)$ is an even function

$$= 0,$$

if $f(x)$ is an odd function.

$$\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx,$$

if $f(2a-x) = f(x)$

if $f(2a-x) = -f(x)$.

$$(x) \text{ Rule of Integration by parts : } \int u(x) v(x) dx = uv_1 - \int u'v_1 dx$$

$$(xi) \text{ Leibnitz General Rule of Integration by parts : } \int u(x) v(x) dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots$$

where dashes denote differentiation of u and number of subscripts denotes number of times v is integrated.